

Project Hubble: Multi-User XR Collaboration Tool

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ABSTRACT

The work introduces Project Hubble, Unity's internal multi-user XR collaboration tool.

Hubble provides a delightful interface to navigate and manipulate freeform and guided experiences. Tools and technology are provided on top for voice chat, avatars, annotations, and sharing.

CCS CONCEPTS

• **human-centered-computing**; • **Interaction design**; • **Human computer interaction**; • **Collaborative and social computing**;

KEYWORDS

Spatial, collaboration, virtual reality

ACM Reference Format:

Brandon Kruysman, Yinghua Yang, Kyle Vaidyanathan, Andrew Maneri, Donnavon Webb, and Matt Schoen. 2022. Project Hubble: Multi-User XR Collaboration Tool. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '22 Immersive Pavilion)*, August 07–11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532834.3536215>

1 INTRODUCTION

Virtual reality usage is increasing and the workplace is becoming more and more distributed. The remote workplace makes it challenging for teams to stay connected, especially when working with real-time 3d content. XR allows for teams to come together in a shared virtual environment regardless of physical location. Project Hubble is an internal tool that allows users to collaboratively author, edit, and review their creations in XR.

Hubble enables users to take a 'virtual field trip' with their teammates, with an interface to create guided walkthrough experiences and networked tools for seamless communication. Multiplayer syncing of voice chat and avatar representation produces an increased feeling of co-presence, promoting entire teams to work closer together regardless of access to the editor. Immersive authoring enables users to get immediate feedback regarding human factors with in-situ, 1:1 scale creation throughout the authoring process.

Collaborative scenes in Hubble are used as a living document, with persistent edits across the entire team.

1.1 Design and Implementation

The design of Hubble grew out of an internal need from the Unity XR team. We rely on evaluating prototypes in the medium or platform that we are designing for. Prior to the pandemic, this meant passing around headsets in person to evaluate prototypes and specific interaction behaviors of spatial features. The tool has become our virtual design studio, with features being developed and reviewed from within Hubble itself. Previous XR authoring tools developed by Unity such as EditorXR and XRI were used as a foundation for functionality, with an added emphasis on modularity and extensibility of components which allows Hubble to adapt to various multiplayer use cases.

1.1.1 Review features. Project Hubble provides various collaboration and review features to aid in the communication of spatial ideas. The walkthrough system acts as a series of spatial 'bookmarks' for moving users through a curated VR experience in a scene. A walkthrough step consists of the following: a teleport anchor which has a camera pose/view scale, a collider volume to trigger steps when freely locomoting in the scene, and corresponding visuals that turn on/off when the step is active. The walkthrough feature allows a presenter to have a focused discussion with various stakeholders on their team. In addition to the walkthrough system, basic head and hands representation for avatars gives users the ability to gesture and point at objects in their environment. An avatar tag floating above the user displays their name and initials, a customizable color or profile pic, along with audio visual cues to notify others in the environment that they are speaking.

1.1.2 Annotation tools. The annotation tools are meant to enable better communication between team members, and promote both real-time and asynchronous collaboration. The default annotation tools are the drawing, comment, and capture tools. All tools are initiated with the toolswitch menu, a configurable spatial UI enabling the quick selection of tools with limited physical effort. The drawing tool offers three types of annotations; planar drawing for diagrammatic representations such as mocking up UI or drawing text, basic freeform drawing with a pipe brush, and drawing with a convex hull brush for volumetric drawing and greyboxing. The capture tool is used for taking in-app screenshots for sharing and review with teammates. Users are given the option to leave screenshots in the scene in order to communicate particular angles or perspectives. The comment tool allows users to leave spatial text annotations in order to collaborate with teammates asynchronously.

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SIGGRAPH '22 Immersive Pavilion, August 07–11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9369-0/22/08.

<https://doi.org/10.1145/3532834.3536215>

These text annotations also can be associated with assets in the scene, giving teammates better context to particular comments.

1.2 Discussion

Immersive technologies are becoming a key part of companies' collaboration toolkits. While all work phases are seeing an increased use of immersive technology, significant improvements have been seen especially in design/prototyping, and marketing/selling activities. Project Hubble has the potential to amplify communication

across teams, inviting non-developers into the creation process. Use cases range from gaming- allowing 3D artists to better collaborate with team members to build and review virtual worlds, to retail-allowing XR developers to work closer with marketing teams for immersive applications. While most XR authoring tools focus on the design and prototyping phase because of time consuming importing/exporting into game engines, Hubble is tightly coupled to the Unity editor and can create a layer of immersive authoring to aid in the entire development process.